

KOTA, India

WMO: 424520

Lat: 25.160N

Lon: 75.846E

Elev: 273

StdP: 98.09

Time Zone: 5.50 (IND)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	11.0	-6.2	2.3	33.3	-3.4	2.9	32.5	4.2	24.4	3.2	21.4	0.4	90	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.1	44.1	20.8	42.9	20.7	41.6	20.6	27.6	31.4	27.1	31.0	26.7	30.7	2.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	23.0	29.5	26.2	22.4	29.0	25.8	21.9	28.7	90.3	31.3	88.2	31.1	86.4	30.7	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	4.9	4.1	7.5	45.9	1.7	1.4	6.3	46.9	5.3	47.7	4.3	48.5	3.1	49.5		
(5)			5.2	28.9	1.3	1.1	4.3	29.8	3.5	30.4	2.8	31.1	1.8	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	17.2	21.1	27.1	33.1	36.5	34.8	30.3	28.8	29.6	28.8	23.9	19.0		
	DBStd	6.38	2.54	3.04	3.36	2.51	2.19	3.11	2.33	1.47	1.76	1.99	2.26	2.58		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	84	54	8	0	0	0	0	0	0	0	0	0	22		
	CDD10.0	6402	222	312	529	692	823	744	631	582	587	583	417	280		
	CDD18.3	3446	18	86	271	442	564	494	372	324	337	325	167	44		
	CDH23.3	48018	122	693	3266	7060	9853	8139	5022	3868	4322	3980	1407	286		
	CDH26.7	28676	14	213	1729	4772	7381	5758	2606	1519	2076	2063	503	43		
(14) Wind	WSAvg	1.3	0.7	0.9	1.1	1.4	2.1	2.4	2.1	1.7	1.3	0.7	0.5	0.5		
(15) Precipitation	PrecAvg	702	4	4	3	2	9	73	253	234	88	14	1	4		
	PrecMax	1415	40	29	34	20	63	201	676	462	369	117	20	43		
	PrecMin	406	0	0	0	0	0	0	34	24	0	0	0	0		
	PrecStd	278	10	7	7	5	13	59	147	111	76	27	4	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	33.2	39.9	43.6	45.7	45.6	40.7	35.5	37.6	37.8	33.7	29.7		
		MCWB	16.7	17.3	17.8	19.1	20.7	21.8	24.3	25.2	22.7	19.8	17.5	17.3		
	2%	DB	26.2	31.2	37.7	42.0	44.5	43.8	37.7	34.1	36.1	36.6	32.2	27.8		
		MCWB	16.2	17.2	17.4	18.8	20.7	21.7	24.8	25.6	23.4	20.2	17.8	16.4		
	5%	DB	24.6	29.4	35.9	40.8	43.5	42.1	35.9	32.9	35.0	35.4	31.1	26.4		
		MCWB	15.4	16.5	17.4	18.7	20.6	22.2	25.1	25.8	23.5	20.1	17.5	15.8		
	10%	DB	22.9	27.6	34.3	39.6	42.4	40.5	34.4	31.9	33.9	34.3	29.8	24.8		
		MCWB	14.4	16.0	17.2	18.4	20.4	22.7	25.4	25.9	23.9	19.8	17.1	15.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	20.0	21.0	21.9	25.4	28.0	28.0	28.2	25.9	23.4	19.2			
		MCDB	25.5	27.2	30.5	37.9	36.1	33.7	31.8	31.4	31.7	30.7	27.1	25.8		
	2%	WB	17.5	18.9	20.0	21.0	24.4	26.7	27.5	27.1	24.9	21.4	18.0			
		MCDB	23.8	27.2	31.2	36.5	36.5	33.1	31.4	30.8	30.7	29.1	25.5	24.9		
	5%	WB	16.4	17.9	19.1	20.3	23.6	26.2	27.1	27.1	26.4	24.0	19.6	16.9		
		MCDB	22.1	26.2	30.7	35.8	37.0	32.9	31.2	30.4	30.2	28.9	26.7	24.0		
	10%	WB	15.4	17.1	18.3	19.7	22.9	25.8	26.8	26.7	25.9	22.9	18.5	16.1		
		MCDB	21.0	25.4	30.6	35.4	37.1	32.8	30.8	29.8	29.8	29.6	27.1	23.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.9	11.9	12.2	11.1	9.2	6.1	5.3	7.2	11.0	11.4	10.9		
		MCDBR	12.2	12.6	13.4	13.1	11.7	10.4	8.4	7.1	9.0	11.8	12.3	12.3		
		MCWBR	5.3	4.7	4.2	3.8	3.4	3.2	2.1	2.1	2.7	3.3	3.7	4.4		
	5% WB	MCDBR	10.7	11.0	11.6	12.0	10.9	8.7	6.3	5.7	6.6	8.2	9.6	10.7		
		MCWBR	5.4	5.0	4.6	4.0	3.8	2.8	2.5	2.3	2.8	3.1	3.9	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.443	0.434	0.470	0.536	0.591	0.666	0.674	0.608	0.539	0.526	0.531	0.487		
	taud		2.011	2.085	2.004	1.879	1.802	1.700	1.708	1.863	1.998	1.918	1.844	1.906		
	Ebn at Noon		792	840	830	783	738	679	674	719	760	741	697	723		
	Edh at Noon		155	154	176	204	220	242	240	205	175	180	182	166		
(44) All-Sky Solar Radiation	RadAvg		4.08	5.11	6.25	7.04	7.17	6.18	4.57	4.40	5.29	5.32	4.34	3.91		
	RadStd		0.32	0.20	0.22	0.20	0.19	0.42	0.55	0.44	0.46	0.25	0.30	0.18		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.17	N/A	N/A	N/A	N/A

Nomenclature: See separate page